



INTERNATIONAL GRAMMAR SCHOOL
Concordia per Divercitatem

Name:

INTERNATIONAL GRAMMAR SCHOOL

MATHEMATICS

YEAR 12

TRIAL EXAMINATION

JULY, 2000

3 UNIT

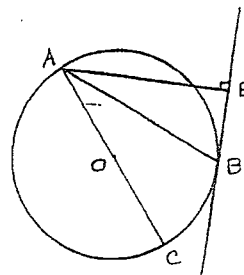
Time allowed — 2 hours
(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page. Number each question clearly.
- Label each page with your name.
- A table of Standard Integrals is attached.

Marks

- Q1. (a) Let A(-5,12) and B(4,9) be two points in the number plane. Find the coordinates of P which divides the interval AB externally in the ratio 5 : 2. 2
- (b) Find the size of the acute angle between the lines $y = 2x + 3$ and $y = 4x + 1$. (Answer to the nearest minute). 2
- (c) Express $f(x) = x^3 + 3x^2 - 10x - 24$ as a product of three linear factors. 3
- (d) Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ 3
- (e) Two points A and B are placed on a circle and AC is a diameter. AE is perpendicular to the tangent at B. 2



- (i) Draw the diagram on your paper.
- (ii) Prove AB bisects $\angle CAE$.

Q2. Start a new booklet

- (a) Solve for x : $x \geq \frac{4}{x}$ 3
- (b) For $y = -3\sin^{-1} \frac{x}{2}$
- (i) State the domain and range.
- (ii) Sketch the curve. 3
- (c) Using the substitution $u = 9 - x^2$, evaluate $\int_0^3 x\sqrt{9-x^2} dx$ 3
- (d) The area bounded by the curve $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x-axis. Find the volume of the solid of revolution. 3

Marks

Q3. Start a new booklet

- (a) Express $3\cos x + 4\sin x$ in the form $A\cos(x-\alpha)$ where $A > 0$. Hence, or otherwise, solve $3\cos x + 4\sin x = -3$ for $0 \leq x \leq 360^\circ$.

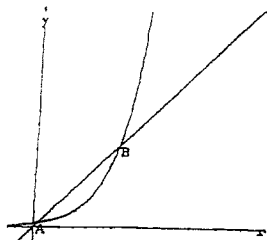
4

- (b) In a co-educational class there are 4 girls and 7 boys. Their classroom has 5 rows of 5 desks neatly arranged. Each student occupies a desk with a chair. Find the number of seating arrangements possible if,

- students can sit anywhere,
- all the girls want to occupy the first row.
- Two particular girls and three particular boys fill the back row seated alternately.

4

(c)



The diagram shows the graphs of $y = e^{x-2}$ and $y = x$ with points of intersection at A and B.

- How many roots has the equation $e^{x-2} - x = 0$?
- Taking $x = 3.3$ as the first approximation, use one application of Newton's Method to find a better approximation to the x-coordinate of B.

4

Marks

Q4. Start a new booklet

- (a) Find x and y if $\frac{4x}{16} = 8^{x+y}$ and $2^{2x+y} = 128$.

3

- (b) If $x = 2 - \cos t$ and $y = 2t + 2\sin t$,

4

- (i) find $\frac{dx}{dt}$ and $\frac{dy}{dt}$

- (ii) Hence or otherwise, find $\frac{dy}{dx}$ in terms of $\frac{t}{2}$.

- (c) A particle is oscillating in simple harmonic motion such that its displacement x metres from the origin is given by the equation $\frac{d^2x}{dt^2} = -9x$ where t is time in seconds.

5

- Show that $x = a \cos(3t + \alpha)$ is a solution of motion for this particle (a and α are constants).
- When $t = 0$, $v = 3$ m/s and $x = 5$ m. Show that the amplitude of the oscillation is $\sqrt{26}$ metres.
- What is the maximum speed of the particle?

Marks

5. Start a new booklet

- (a) α, β, γ are the roots of the equation $2x^3 + 3x^2 - 4 = 0$

3

Find

(i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta\gamma$

(iii) $\alpha^2 + \beta^2 + \gamma^2$

- (b) For the function $y = x^2 - 2x + 1$, find the largest possible domain such that this function has an inverse. Find the equation of this inverse and state its range.

3

- (c) For the parabola $x^2 = 12y$, find

6

- (i) the equation of the tangent at the point $P(6p, 3p^2)$ on the parabola.

- (ii) the coordinates of the point T where the tangent meets the x axis.

- (iii) Show that N , the midpoint of PT , has coordinates $(\frac{9p}{2}, \frac{3p^2}{2})$.

- (iv) Find the equation of the locus of N .

Start a new booklet

- (a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

2

- (b) The daily growth of a colony of insects is 10% of the excess of the population over 1.2×10^6 .

4

ie $\frac{dN}{dt} = 0.1(N - 1.2 \times 10^6)$.

Initially, the population is 2.7×10^6 .

- (i) Determine the population after $3\frac{1}{2}$ days.

- (ii) If a scientist checks the population each day, which is the first day on which she should notice that the original population has tripled?

Marks

Q6. (continued).....

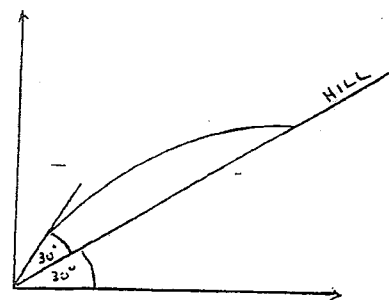
- (c) A ball is thrown with a velocity of $30\sqrt{3}$ m/s at an angle of 60° to the horizontal.

6

- (i) Assuming negligible air resistance and letting $g = 10 \text{ ms}^{-2}$, derive the equations of motion.

- (ii) Find the time of flight and the range.

- (iii) If the ball had been thrown with velocity $30\sqrt{3}$ m/s at an angle of 30° to a hill which is itself inclined at 30° to the horizontal (see diagram), determine the time of flight.



Q 7. Start a new booklet

- (a) Prove by mathematical induction that for all values of n

5

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

where n is a positive integer.

- (b) (i) Show that $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ has no stationary points.

7

- (ii) Prove that the lines $y = \pm 1$ are asymptotes.

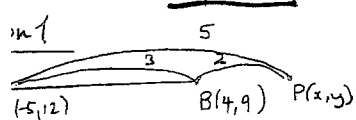
- (iii) Sketch the curve.

- (iv) If k is a positive constant, find the area in the first quadrant enclosed by the above curve and the three lines $y = 1$, $x = 0$ and $x = k$.

- (v) Prove that for all values of k , this area is always less than $\log_e 2$.

11. a.

it Trial July 2000



$$\frac{3x+2(-5)}{5}, \quad 9 = \frac{3y+2(12)}{5}$$

$$\begin{aligned} 3x-10 &= 45=3y+24 \\ 3x &= 21=3y \\ 10 &= 7 \end{aligned}$$

$$P = (10, 7)$$

$$= 2, m_2 = 4.$$

$$\begin{aligned} m &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{2 - 4}{1 + 8} \right| \end{aligned}$$

$$m = \frac{2}{9}$$

$$\alpha = 12^\circ 32'$$

$$f(x) = x^3 + 3x^2 - 10x - 24$$

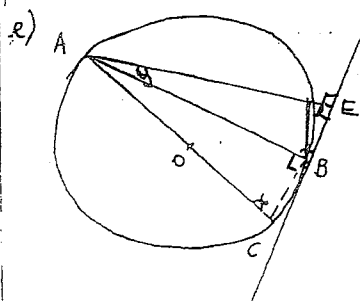
$$f(3) = 27 + 27 - 30 - 24 = 0$$

3) is a factor

$$\begin{array}{r} x^3 + 3x^2 - 10x - 24 \\ x-3 \overline{) } \\ \underline{x^3 - 3x^2} \\ 6x^2 - 10x \\ \underline{6x^2 - 18x} \\ 8x - 24 \\ \underline{8x - 24} \\ 0 \end{array}$$

$$\begin{aligned} f(x) &= (x-3)(x^2+6x+8) \\ f(x) &= (x-3)(x+2)(x+4) \end{aligned}$$

$$\begin{aligned} d) \int_0^3 \frac{dx}{\sqrt{9-x^2}} &= \left[\sin^{-1} \frac{x}{3} \right]_0^3 \\ &= \sin^{-1} 1 - \sin^{-1} 0 \\ &= \frac{\pi}{2} \end{aligned}$$



Construction: join BC

$\angle ABC = 90^\circ$ (angle in semicircle)

$\angle AEB = 90^\circ$ (adj. suppl. $\angle$)

let $\angle EAB = y$, $\angle ABE = x$

then $x + y = 90^\circ$ (angle sum of Δ)

$\angle ACB = x$ (angle in alt segment)

$\therefore \angle CAB = 180 - 90 - x$

$= 90 - x$ (angle sum of Δ)

$= y$

$\therefore \angle CAB = \angle BAE = y$

AB bisects $\angle CAE$

Question 2

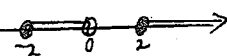
$$a) x \geq \frac{4}{x}$$

$x \neq 0$;

$$\text{solve } x^2 = 4$$

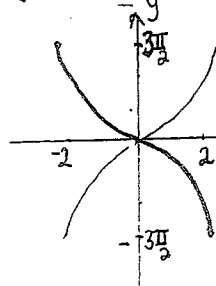
$$x = \pm 2$$

Try solutions



$$\therefore -2 \leq x < 0 \text{ or } x \geq 2$$

$$b) y = -3 \sin^{-1} \frac{x}{2}$$



Domain: $-2 \leq x \leq 2$

Range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

$$c) u = 9 - x^2 \quad \text{limits}$$

$$\frac{du}{dx} = -2x \quad \text{when } x=0, u=9$$

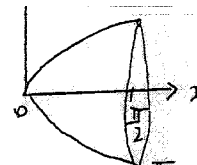
$$\frac{du}{dx} = -2x \quad \text{when } x=3, u=0$$

$$\int_0^3 x \sqrt{9-x^2} dx = -\frac{1}{2} \int_9^0 \sqrt{u} du$$

$$= \frac{1}{2} \int_0^9 u^{1/2} du$$

$$= \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_0^9$$

d)



$$V = \pi \int_0^{\pi/2} \sin^2 x dx$$

$$= \frac{\pi}{2} \int_0^{\pi/2} 1 - \cos 2x dx$$

$$= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} ((\frac{\pi}{2} - 0) - (0 - 0))$$

$$= \frac{\pi^2}{4} \text{ units}^3$$

on 3

$$s(x-\alpha) = 3\cos x + 4\sin x$$

$$2s\alpha + 1\sin x \sin \alpha = 3\cos x + 4\sin x$$

$$t\cos \alpha = 3 \dots \textcircled{1}$$

$$\sin \alpha = 4 \dots \textcircled{2}$$

$$\tan \alpha = \frac{4}{3}$$

$$\alpha = 53^\circ$$

$$A^2(\cos^2 \alpha + \sin^2 \alpha) = 3^2 + 4^2$$

$$A^2 = 25$$

$$A = 5$$

$$5x + 4\sin x = 5\cos(x-53^\circ)$$

$$5\cos(x-53^\circ) = -3$$

$$\cos(x-53^\circ) = -\frac{3}{5}$$

$$x-53^\circ = 127^\circ, 233^\circ$$

$$x = 180^\circ, 286^\circ$$

IRLS, 7 BOYS.

x x x x

x x x x

x x x x

x x x x

"

7 x 5P₄

6 x ³P₃ x ²P₂

c) (i) 2 roots

$$(ii) f(x) = e^{x-2} - x$$

$$f'(x) = e^{x-2} - 1$$

$$a_2 = 3.3 - \frac{f(3.3)}{f'(3.3)}$$

$$= 3.3 - \frac{0.369}{2.669}$$

$$= 3.16$$

Question 4

$$a) \frac{4^x}{16} = 8^{x+y} \text{ and } 2^{2x+y} = 128$$

change all to powers of 2

$$2^{2x-4} = 2^{3x+3y}$$

$$2x-4 = 3x+3y \text{ (equating indices)}$$

$$-x-3y = 4 \dots \textcircled{1}$$

$$2^{2x+y} = 2^7$$

$$2x+y = 7 \dots \textcircled{2} \text{ (equating indices)}$$

solving $\textcircled{1}$ & $\textcircled{2} \Rightarrow$

$$-2x+6y = 8$$

$$+ \quad 2x+y = 7$$

$$-5y = 15$$

$$y = -3 \therefore x = 5$$

$$b) x = 2 - \cos t \quad y = 2t + 2\sin t$$

$$\frac{dx}{dt} = +\sin t$$

$$\frac{dy}{dt} = 2 + 2\cos t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{2+2\cos t}{\sin t}$$

$$\text{where } \cos t = \frac{1-t^2}{1+t^2}$$

$$\sin t = \frac{2t}{1+t^2}$$

$$\frac{dy}{dx} = \frac{2 + 2\left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}}$$

$$v^2 = 9(26-0) \therefore v = \pm 3\sqrt{26} \therefore \text{max speed is } 3\sqrt{2} \text{ m/s}$$

$$\frac{dy}{dx} = \frac{2(1+t^2) + 2(1-t^2)}{2t}$$

$$= \frac{2+2t^2 + 2-2t^2}{t}$$

$$= \frac{4}{t}$$

$$c) x = a \cos(3t+\alpha)$$

$$\dot{x} = -3a \sin(3t+\alpha)$$

$$\ddot{x} = -9a \cos(3t+\alpha)$$

$$\ddot{x} = -9(a \cos(3t+\alpha))$$

$$\ddot{x} = -9x$$

$\therefore$ satisfies the ..

when $t=0$ and $x=5$

using $\textcircled{1}$

$$5 = a \cos \alpha$$

using $\textcircled{2}$

$$3 = -3a \sin \alpha$$

$$-1 = a \sin \alpha$$

$$\textcircled{3}^2 + \textcircled{4}^2 \Rightarrow 25 + 1 = a^2 \therefore a = \sqrt{26}$$

$$\textcircled{4} \div \textcircled{3} \Rightarrow \tan \alpha = -1/5$$

$$\alpha = -0.11$$

(iii) max speed at centre

$$\text{From } \textcircled{1} \quad 0 = \sqrt{26} \cos(3t-0.11)$$

$$3t-0.11 = \frac{\pi}{2}$$

$$\therefore v = -3\sqrt{26} \sin(3(0.56)-0.11)$$

on 5

$$\alpha + \beta + \gamma = -\frac{3}{2}$$

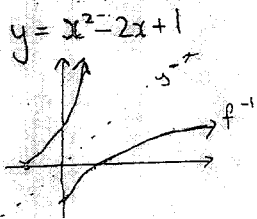
$$\alpha\beta\gamma = -(-\frac{3}{2}) = 2$$

$$\beta + \gamma + \alpha\gamma = 0$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= (-\frac{3}{2})^2$$

$$= \frac{9}{4}$$



gest domain of f : $x \geq -1$

let $y = x^2 - 2x + 1$

ange $x = y^2 - 2y + 1$

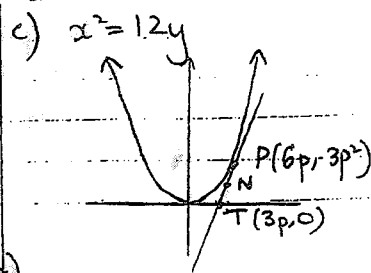
ke y the subject

$$x = (y-1)^2$$

$$\sqrt{x} = y-1$$

$$y = \sqrt{x} + 1$$

nge: $y \geq -1$



a) $\frac{x^2}{12} = y$

$$\frac{dy}{dx} = \frac{2x}{12} = \frac{x}{6}$$

at $x = 6p$, $\frac{dy}{dx} = \frac{6p}{6} = p$

$\therefore$ eqn of tangent is

$$(y - 3p^2) = p(x - 6p)$$

$$y - 3p^2 = px - 6p^2$$

$$y = px - 3p^2$$

(i) tangent meets x -axis when $y=0$

$$px = 3p^2$$

$$x = 3p$$

$$\therefore T(3p, 0)$$

(ii) N is mid pt of PT

$$N \equiv \left(\frac{6p+3p}{2}, \frac{3p^2}{2} \right)$$

$$N \equiv \left(\frac{9p}{2}, \frac{3p^2}{2} \right)$$

(iv) let $X = \frac{9p}{2}$, $Y = \frac{3p^2}{2}$

$$X^2 = \frac{81p^2}{4}$$

$$X^2 = \frac{27}{2} \left(\frac{3p^2}{2} \right) = \frac{27}{2} Y$$

Question 6

a) $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \times \frac{3}{5} \right)$

$$= \frac{3}{5} (1)$$

$$= \frac{3}{5}$$

b) $\frac{dN}{dt} = 0.1(N - 1.2 \times 10^6)$

Solution is

$$N = A + Be^{kt}$$

where $A = 1.2 \times 10^6$

$$k = 0.1$$

when $t=0$, $N = 2.7 \times 10^6$

$$2.7 \times 10^6 = 1.2 \times 10^6 + Be^0$$

$$B = 1.5 \times 10^6$$

$$N = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.1t}$$

(i) when $t = 3\frac{1}{2}$

$$N = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.35}$$

$$N = 3328601$$

(ii) when $N = 2.7 \times 10^6 \times 3$

$$N = 8.1 \times 10^6$$

$$8.1 \times 10^6 = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.1t}$$

$$\frac{6.9 \times 10^6}{1.5 \times 10^6} = e^{0.1t}$$

$$\ln 4.6 = t \quad t = 15.26$$

c) (i) $\ddot{x} = 0$ $\ddot{y} = -10$

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = -10t + V \sin \alpha$$

$$\dot{x} = 30\sqrt{3} \cos 60^\circ$$

$$\dot{y} = -10t + 30\sqrt{3} \sin 60^\circ$$

$$\dot{x} = 15\sqrt{3}$$

$$\dot{y} = -10t + 45$$

$$x = 15\sqrt{3}t + 0$$

$$y = -5t^2 + 45t$$

(ii) For time of flight

when $y=0$, $5t^2 - 45t = 0$

$$5t(t-9) = 0$$

$$t=0, t=9$$

9 seconds

For range;

$$x = 15\sqrt{3} \times 9 = 135\sqrt{3} \text{ m}$$

(iii) equation of hill, $y = \frac{1}{\sqrt{3}}x$

$$-5t^2 + 45t = \frac{1}{\sqrt{3}}(15\sqrt{3})t$$

$$-5t^2 + 30t = 0$$

$$-5t(t-6) = 0$$

$$t=0 \text{ or } t=6$$

$\therefore$ time of flight is 6 seconds

lim 7

r n=1

$$s = \frac{1}{2!} = \frac{1}{2}$$

$$= \frac{(1+1)! - 1}{2!} = \frac{2-1}{2} = \frac{1}{2}$$

me for n=1

same true for n=k

$$\frac{2}{3!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!}$$

for n=k+1

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$\frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)[(k+1)! - 1] + (k+1)}{(k+2)!}$$

$$= \frac{(k+2)! - (k+2) + (k+1)}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$\frac{(k+2)! - 1}{(k+2)!}$$

true for n=k+1

true for n=1

true for all n, positive

$$b) (i) y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\frac{dy}{dx} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{(e^x + e^{-x})^2}$$

$$\frac{dy}{dx} = \frac{4}{(e^x + e^{-x})^2}$$

but $\frac{dy}{dx} \neq 0$ no statpts

as $4 \neq 0$ and $e^x + e^{-x} \neq 0$

$$(ii) y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\lim_{x \rightarrow \infty} \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)$$

$$\lim_{x \rightarrow \infty} \left(\frac{\frac{e^x}{e^x} - \frac{e^{-x}}{e^x}}{\frac{e^x}{e^x} + \frac{e^{-x}}{e^x}} \right)$$

$$= \left(\frac{1-0}{1+0} \right) = 1$$

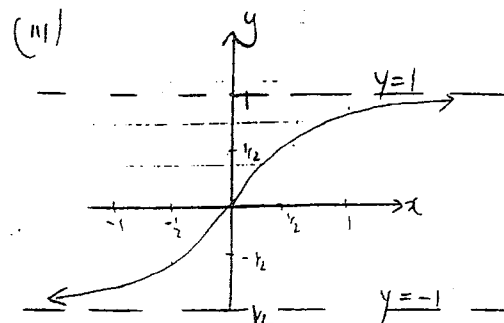
$y=1$ is an asymptote

$$\text{Similarly } \lim_{x \rightarrow -\infty} \left(\frac{\frac{e^x}{e^{-x}} - \frac{e^{-x}}{e^{-x}}}{\frac{e^x}{e^{-x}} + \frac{e^{-x}}{e^{-x}}} \right)$$

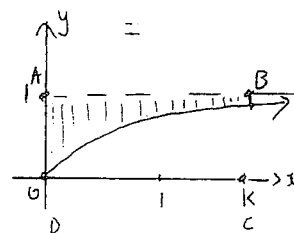
$$= \frac{-1}{-1} = 1$$

$y=-1$ is an asymptote

(iii)



(iv)



Area of rectangle ABCD = $k \times 1 = k$

$$\begin{aligned} \text{Area under curve} &= \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx \\ &= \ln(e^x + e^{-x}) \Big|_0^k \\ &= \ln(e^k + e^{-k}) - \ln 2 \end{aligned}$$

$$\therefore \text{Shaded area} = k - (\ln(e^k + e^{-k}) - \ln 2)$$

$$(v) \text{ Area} = k - \ln(e^k + e^{-k}) + \ln 2$$

$$\begin{aligned} &= \log_e e^k - \ln(e^k + e^{-k}) + \ln 2 \\ &= \log_e \frac{e^k}{e^k + e^{-k}} + \ln 2 \end{aligned}$$

$$\text{Now, } \log \left[\frac{e^k}{e^k + e^{-k}} \right] < 0$$

$$\text{as } \frac{e^k}{e^k + e^{-k}} < 1$$

$\therefore$ max value is $\log_e 2$